

Two-Wheeler Driving Behaviour Recognition

Comprehensive Model Report

Model explanations, hyperparameters, metrics, graphs, confusion matrices, and final comparison

1. Executive Summary

The strongest measured classification result is histgradientboosting with accuracy 94.60% and macro F1 0.9461. The current benchmark evaluates 500 windows from the full window artifact because only 209 random-test windows are available; therefore, these values are comparative engineering measurements, not independent generalisation estimates.

MAE is reported on integer-encoded labels as a diagnostic only. Since the behaviour classes are nominal, macro F1 is the primary balanced metric.

2. Dataset and Metrics

Item	Value
Features	Ax, Ay, Az, Gx, Gy, Gz, Speed
Window	156 samples
Stride	78 samples
Classes	STRAIGHT, LEFT, RIGHT, BUMP, STOP
Accuracy	Correct predictions / total predictions
Macro F1	Unweighted mean of the five class F1 scores
MAE	Mean absolute error on encoded class IDs; diagnostic only

3. Model Explanations and Hyperparameters

Model	Family	What it does	Hyperparameters
logistic_regression	Linear baseline	Multiclass linear classifier over flattened 156x7 windows; interpretable and fast.	max_iter=500, class_weight=balanced, solver=lbfgs
linear_svm	Linear baseline	Linear maximum-margin classifier over flattened sensor windows.	class_weight=balanced, max_iter=3000
rbf_svm	Kernel baseline	Nonlinear RBF-kernel support-vector classifier for flattened windows.	kernel=rbf, probability=True, class_weight=balanced
random_forest	Tree ensemble	Bagged decision trees that model nonlinear feature interactions.	n_estimators=100, class_weight=balanced, n_jobs=-1
histgradientboosting	Tree boosting	Histogram-based gradient boosting over flattened sensor features.	n_estimators-style boosting, max_iter=100, random_state=42
lightgbm	Tree boosting	LightGBM multiclass gradient boosting optimized for tabular data.	n_estimators=200, learning_rate=0.05, num_leaves=31, class_weight=balanced
xgboost	Tree boosting	Regularized gradient-boosted trees used as the deployment baseline.	n_estimators=200, max_depth=6, learning_rate=0.05, subsample=0.9, colsample_bytree=0.8
mlp_tabular	Feed-forward neural network	Two-hidden-layer Keras MLP over flattened windows.	Dense=128, Dense=64, dropout=0.2, Adam, batch_size=64, epochs=10
gru	Recurrent neural network	Two-layer gated recurrent sequence model for temporal dependencies.	GRU=64 return_sequences, dropout=0.2, GRU=32, Adam, batch_size=64, max_epochs=30
lstm	Recurrent neural network	Single-layer LSTM HAR classifier.	LSTM=64, dropout=0.2, Adam, batch_size=64, max_epochs=30
resnet1d	Convolutional neural network	Residual 1D temporal convolution model exported to ONNX.	input channels=7, base channels=16, residual blocks=2, Adam, epochs=10
convnet_har	Convolutional neural network	Three-block 1D ConvNet with global	Conv1D filters=64, kernels=7/5/3, global

		average pooling.	average pooling, Adam, epochs=10
sgconv	Structured convolution	Compact depthwise and dilated convolution approximation of SGConv.	channels=32/64, dilation=2/4, GELU, Adam, epochs=10
s4d	State-space model	Compact diagonal state-space recurrence exported to ONNX.	state_dim=32, learned decay, LayerNorm, Adam, epochs=10
s5	State-space model	Existing Keras/TFLite S5-style recurrent state-space classifier.	state_size=16, sigmoid decay, Adam, epochs=12
s5_onnx	State-space model	ONNX state-space variant evaluated alongside S4D and LRU.	state_dim=32, learned decay, LayerNorm, Adam, epochs=10
lru	State-space model	Linear recurrent unit approximation with learned phase modulation.	state_dim=32, learned phase and decay, Adam, epochs=10
tst	Transformer	Compact Temporal-Spatial Transformer with positional embeddings.	d_model=24, heads=2, feedforward=48, blocks=2, dropout=0.1, epochs=12
tcnca_v1	Temporal convolution ablation	Dilated temporal CNN without chunked attention.	channels=24, dilations=[1,2,4,8,16], no attention, epochs=8
tcnca_v2	Temporal convolution ablation	Dilated temporal CNN with chunked attention using 26-sample chunks.	channels=24, chunk_size=26, dilations=[1,2,4,8,16], epochs=8
tcnca_v3	Temporal convolution ablation	Final TCNCA ablation with larger chunks and five dilation rates.	channels=24, chunk_size=52, dilations=[1,2,4,8,16], epochs=8
mega	Transformer/attention	Moving-average gated attention approximation for temporal smoothing and attention.	Conv1D channels=32, average pool=9, heads=4, key_dim=8, epochs=8
fusformer	Transformer/attention	Fusion Transformer combining short and long convolutional branches.	Conv kernels=5/11, fused channels=32, heads=4, epochs=8
conv_transformer	Transformer/attention	Convolutional stem followed by two transformer encoder	Conv1D channels=32, heads=4, blocks=2, feedforward=64, epochs=8

		blocks.	
dtaad	Anomaly detection	Dual dilated-TCN autoencoder with temporal attention; returns novelty score, not a class.	Two TCN branches, dilations=1/2/4, attention heads=2, normal label=STRAIGHT, epochs=15
egodrive_rt	Stub/future	Stub only because pretrained EgoDriveRT weights are not public.	Not applicable
egodrive_max	Stub/rejected	Rejected because the model is expected to exceed 10M parameters and the 10 ms budget.	Not applicable

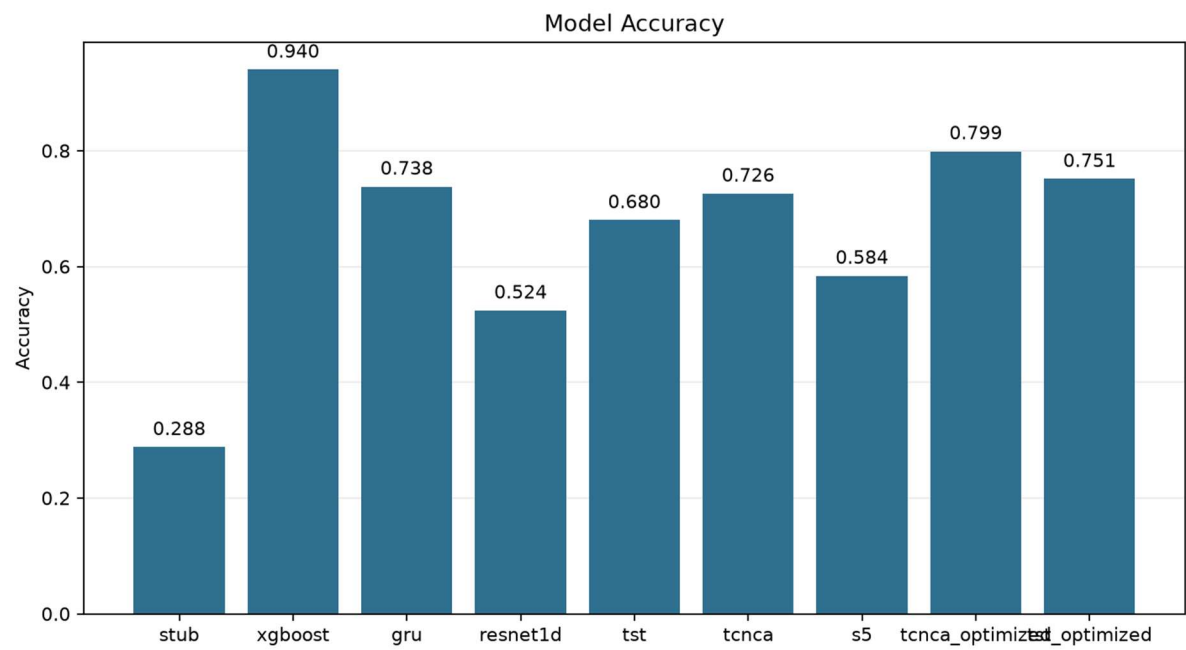
4. Metrics for Every Model

Model	Family	MAE	Accuracy	Macro F1	Mean latency (ms)	P95 latency (ms)	Parameters	Status
histgradientboosting	Tree boosting	0.1680	0.9460	0.9461	76.7518	116.5344	10	benchmarked
lightgbm	Tree boosting	0.1680	0.9440	0.9443	4.6147	5.9961	5	benchmarked
random_forest	Tree ensemble	0.1940	0.9380	0.9415	35.1453	47.5854	5	benchmarked
xgboost	Tree boosting	0.1860	0.9400	0.9412	3.6804	4.7093	0	benchmarked
logistic_regression	Linear baseline	0.3520	0.8740	0.8630	0.1348	0.2273	5471	benchmarked
linear_svm	Linear baseline	0.3980	0.8440	0.8280	0.1028	0.1302	5470	benchmarked
mlp_tabular	Feed-forward neural network	0.8000	0.7480	0.7363	130.5565	272.5350	0	benchmarked
conv_transformer	Transformer/attention	0.7520	0.7520	0.7274	1.1508	1.4676	18853	benchmarked
gru	Recurrent neural network	0.9000	0.7140	0.6828	220.9388	356.3285	0	benchmarked
lstm	Recurrent neural network	0.7920	0.7260	0.6504	157.9293	253.6587	0	benchmarked
fusformer	Transformer/attention	1.0500	0.6740	0.6464	0.5723	0.7482	10533	benchmarked
mega	Transformer/attention	1.0300	0.6620	0.6260	0.6102	0.8412	10917	benchmarked
convnet_har	Convolutional neural network	0.9940	0.6820	0.6023	0.5997	1.2880	36933	benchmarked
tcnca_v3	Temporal convolution ablation	1.2500	0.6040	0.5639	0.2694	0.3501	44261	benchmarked
rbf_svm	Kernel baseline	1.2520	0.5940	0.5635	0.7440	1.5168	741048	benchmarked
tcnca_v2	Temporal convolution ablation	1.2720	0.5860	0.4746	0.2869	0.5039	29285	benchmarked
tcnca_v1	Temporal convolution ablation	1.3340	0.5760	0.4444	0.2512	0.4615	9413	benchmarked
tst	Transformer	1.5020	0.5400	0.3941	0.7138	0.8614	10109	benchmarked
s4d	State-space model	1.6120	0.4980	0.2541	3.0845	4.0810	1829	benchmarked
s5_onnx	State-space model	1.6120	0.4980	0.2541	2.9414	3.9633	1829	benchmarked

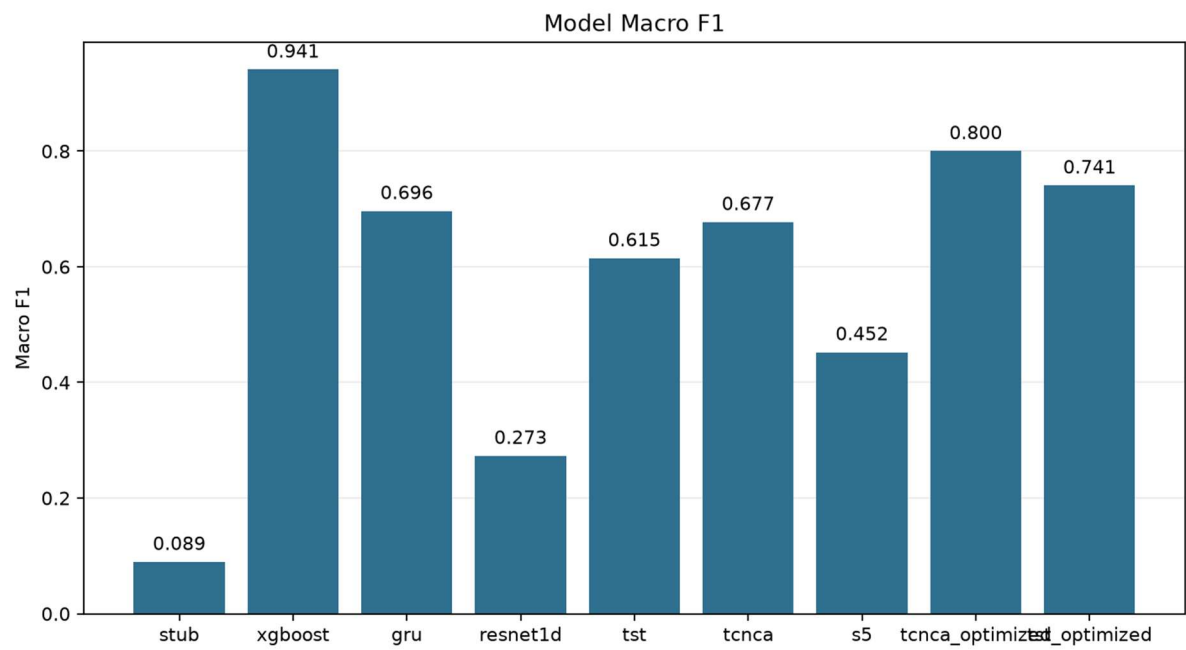
lru	State-space model	1.6120	0.4980	0.2541	3.3156	4.4430	1861	benchmarked
sgconv	Structured convolution	1.6160	0.4820	0.1301	0.4099	1.1704	2437	benchmarked
resnet1d	Convolutional neural network	1.6160	0.4820	0.1301	0.1399	0.3122	3653	benchmarked
egodrive_rt	Stub/future	2.3840	0.2880	0.0894	0.0047	0.0050	0	benchmarked
egodrive_max	Stub/rejected	2.3840	0.2880	0.0894	0.0038	0.0046	0	benchmarked
dtaad	Anomaly detection	N/A	nan	nan	0.5110	0.7205	11399	anomaly_only

DTAAD is anomaly-only and therefore has no classification accuracy, F1, or class-label MAE.

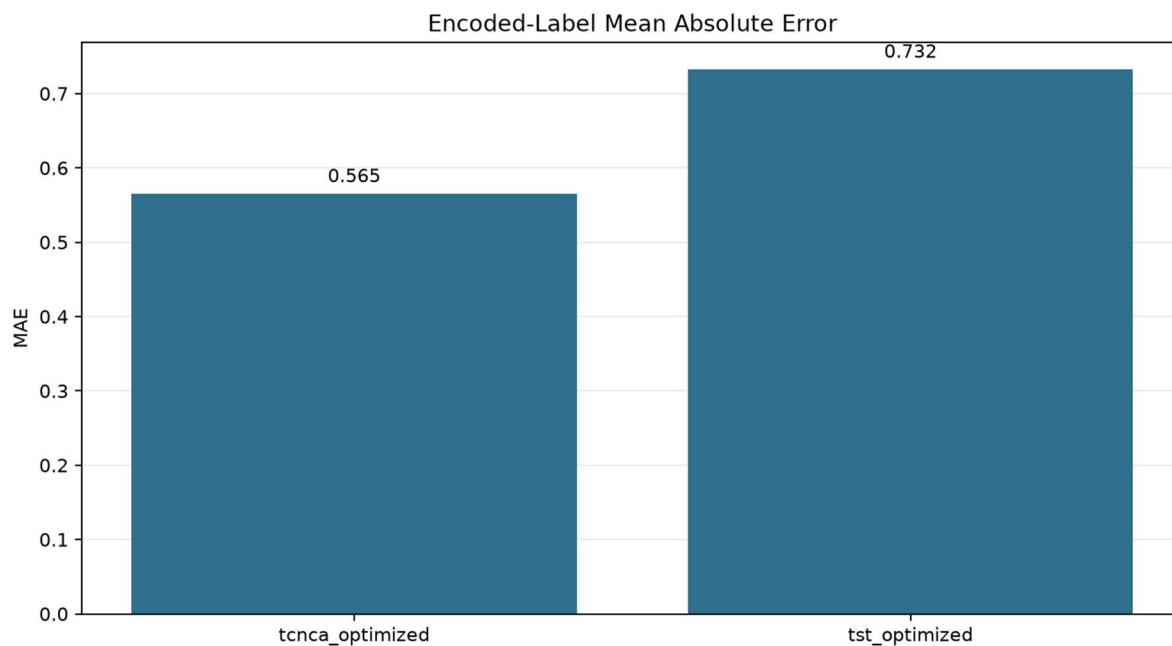
5. Benchmark Graphs



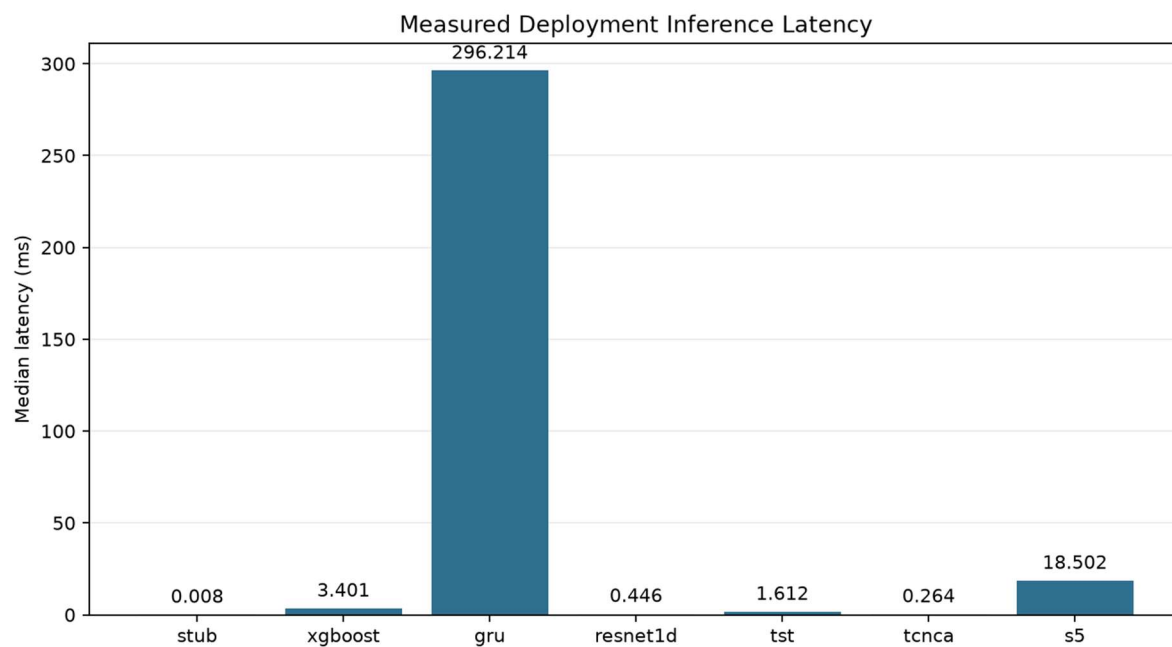
Accuracy comparison



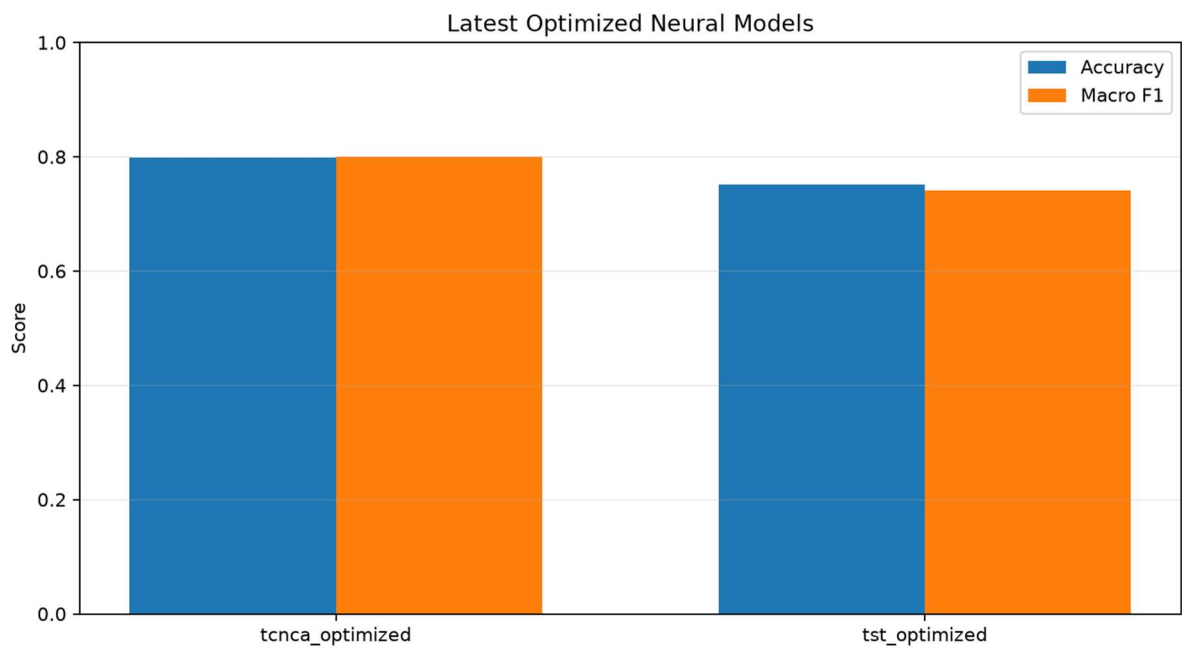
Macro F1 comparison



Encoded-label MAE comparison



Latency comparison

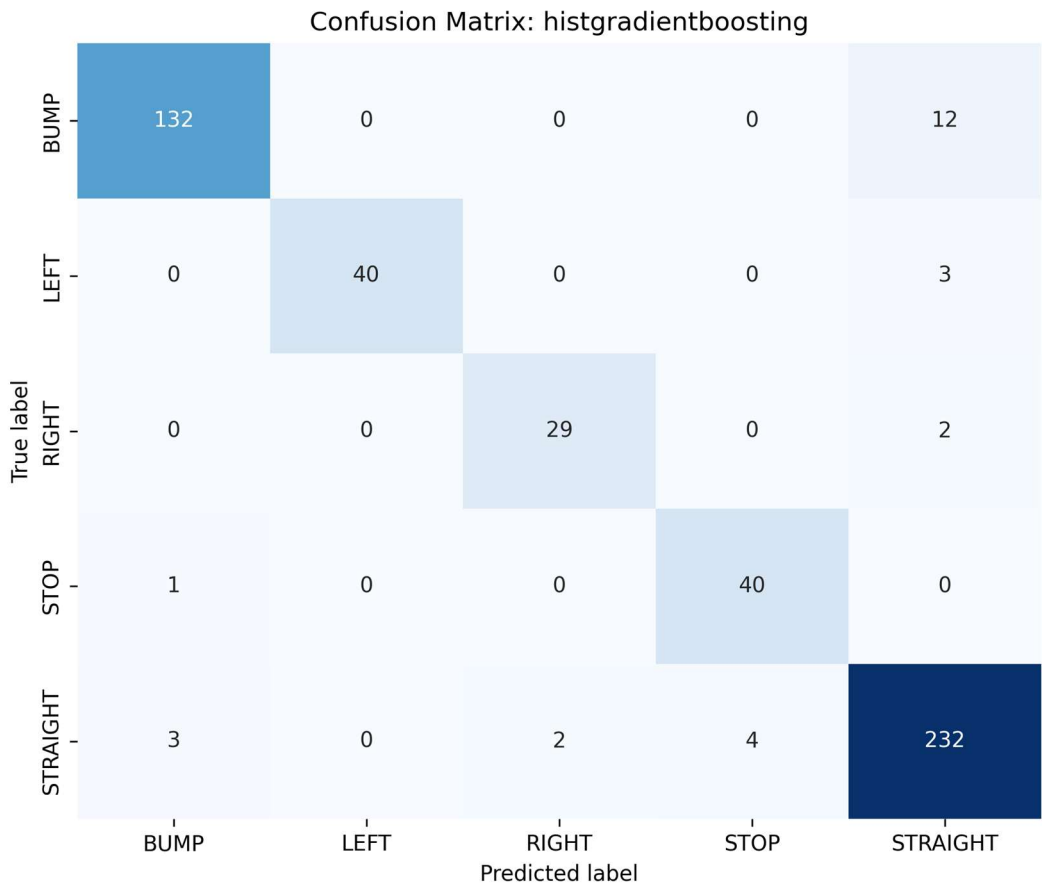


Optimized neural comparison

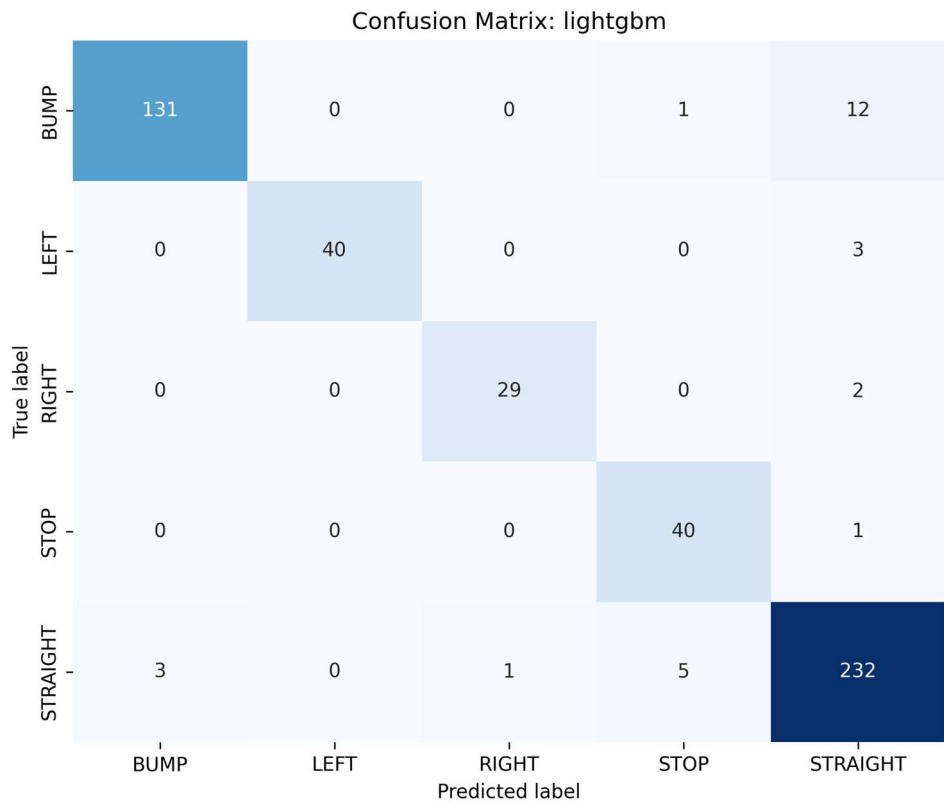
6. Confusion Matrices and Class-Wise Results

Model	Class	Precision	Recall	F1	Support
logistic_regression	BUMP	0.8543	0.8958	0.8746	144
logistic_regression	LEFT	0.8750	0.8140	0.8434	43
logistic_regression	RIGHT	0.9032	0.9032	0.9032	31
logistic_regression	STOP	0.6964	0.9512	0.8041	41
logistic_regression	STRAIGHT	0.9279	0.8548	0.8898	241
logistic_regression	macro avg	0.8514	0.8838	0.8630	500
logistic_regression	weighted avg	0.8817	0.8740	0.8753	500
linear_svm	BUMP	0.8435	0.8611	0.8522	144
linear_svm	LEFT	0.8947	0.7907	0.8395	43
linear_svm	RIGHT	0.9630	0.8387	0.8966	31
linear_svm	STOP	0.5352	0.9268	0.6786	41
linear_svm	STRAIGHT	0.9217	0.8299	0.8734	241
linear_svm	macro avg	0.8316	0.8494	0.8280	500
linear_svm	weighted avg	0.8677	0.8440	0.8498	500
random_forest	BUMP	0.9844	0.8750	0.9265	144
random_forest	LEFT	1.0000	0.9302	0.9639	43
random_forest	RIGHT	1.0000	0.9355	0.9667	31
random_forest	STOP	0.8511	0.9756	0.9091	41
random_forest	STRAIGHT	0.9141	0.9710	0.9416	241
random_forest	macro avg	0.9499	0.9375	0.9415	500
random_forest	weighted avg	0.9419	0.9380	0.9381	500
histgradientboosting	BUMP	0.9706	0.9167	0.9429	144
histgradientboosting	LEFT	1.0000	0.9302	0.9639	43
histgradientboosting	RIGHT	0.9355	0.9355	0.9355	31
histgradientboosting	STOP	0.9091	0.9756	0.9412	41
histgradientboosting	STRAIGHT	0.9317	0.9627	0.9469	241
histgradientboosting	macro avg	0.9494	0.9441	0.9461	500
histgradientboosting	weighted avg	0.9472	0.9460	0.9460	500
lightgbm	BUMP	0.9776	0.9097	0.9424	144
lightgbm	LEFT	1.0000	0.9302	0.9639	43
lightgbm	RIGHT	0.9667	0.9355	0.9508	31
lightgbm	STOP	0.8696	0.9756	0.9195	41
lightgbm	STRAIGHT	0.9280	0.9627	0.9450	241

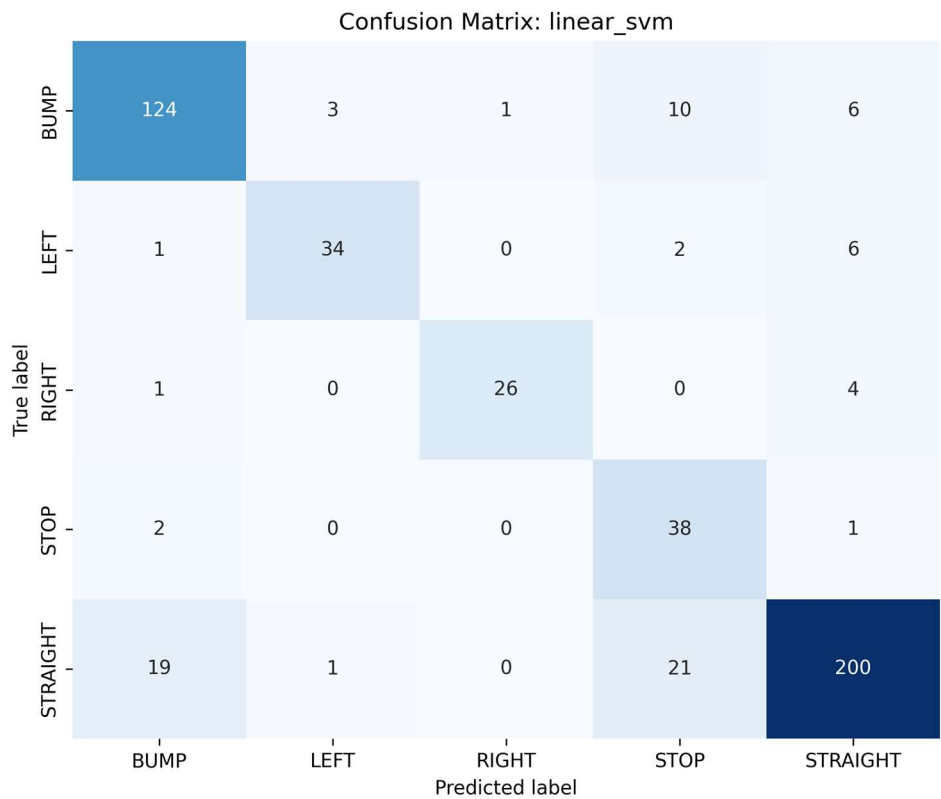
lightgbm	macro avg	0.9484	0.9427	0.9443	500
lightgbm	weighted avg	0.9461	0.9440	0.9442	500
xgboost	BUMP	0.9565	0.9167	0.9362	144
xgboost	LEFT	1.0000	0.9302	0.9639	43
xgboost	RIGHT	0.9355	0.9355	0.9355	31
xgboost	STOP	0.8889	0.9756	0.9302	41
xgboost	STRAIGHT	0.9309	0.9502	0.9405	241
xgboost	macro avg	0.9424	0.9416	0.9412	500
xgboost	weighted avg	0.9411	0.9400	0.9401	500



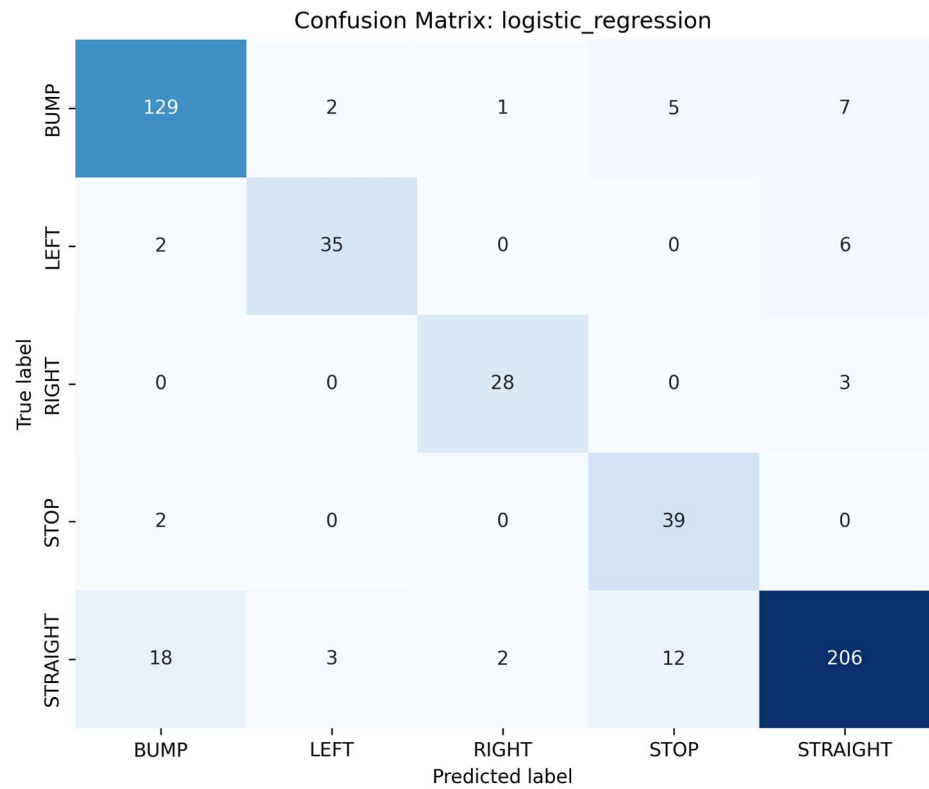
Confusion matrix: histgradientboosting



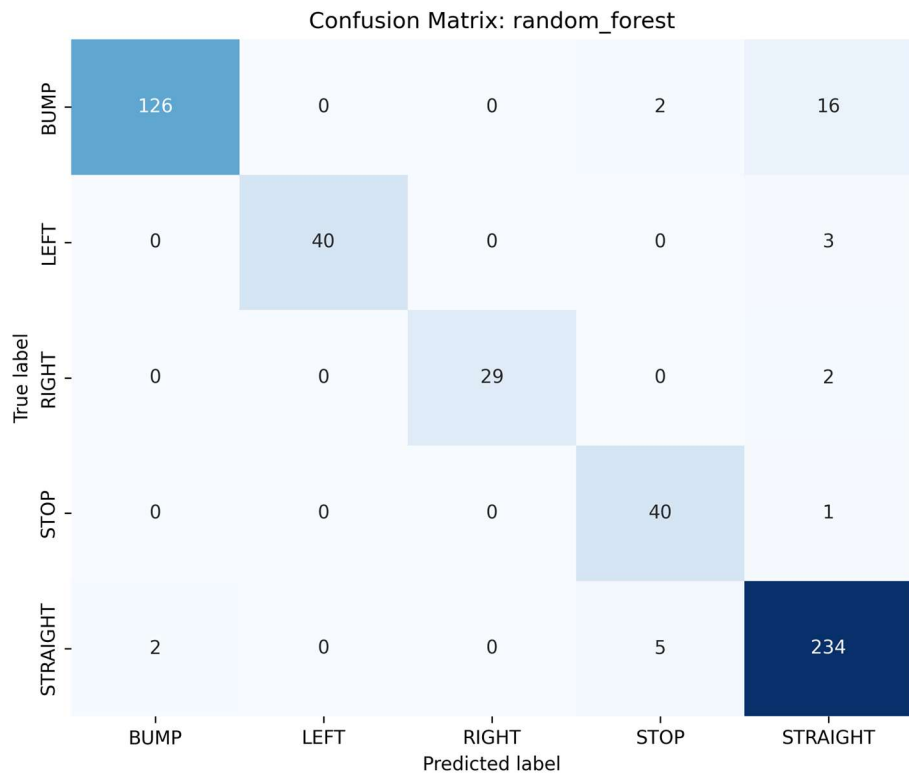
Confusion matrix: lightgbm



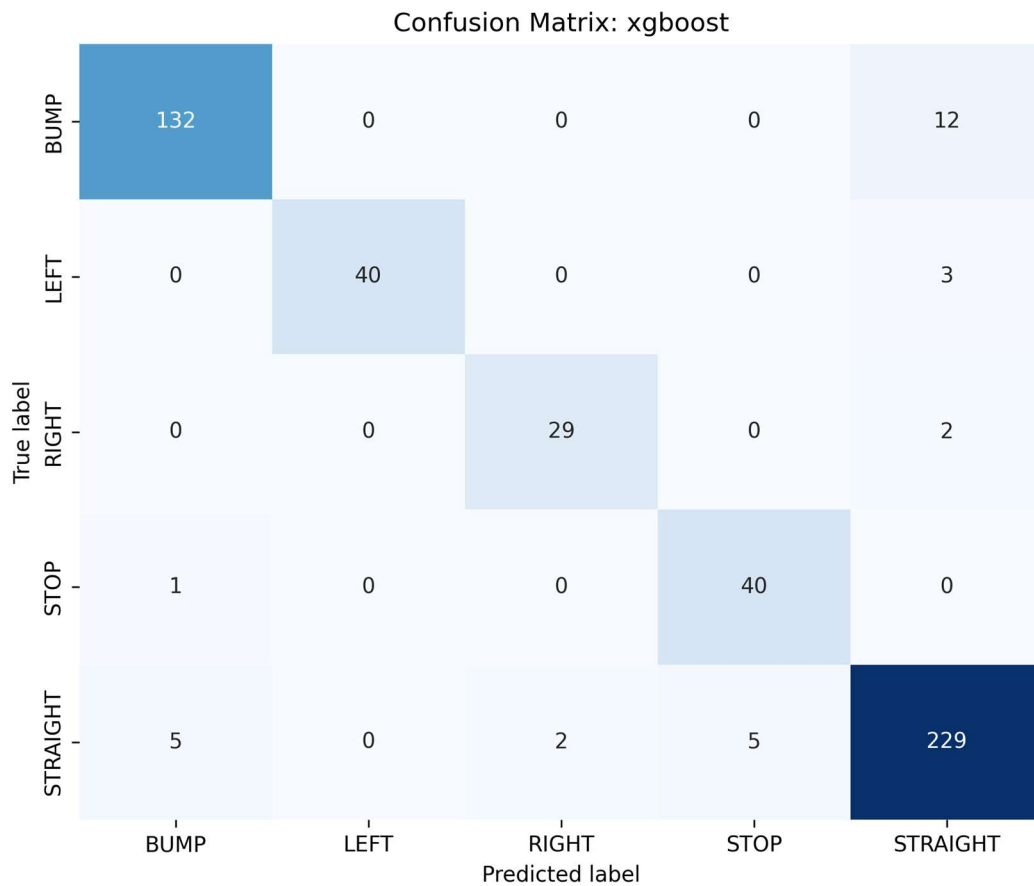
Confusion matrix: linear_svm



Confusion matrix: logistic_regression



Confusion matrix: random_forest



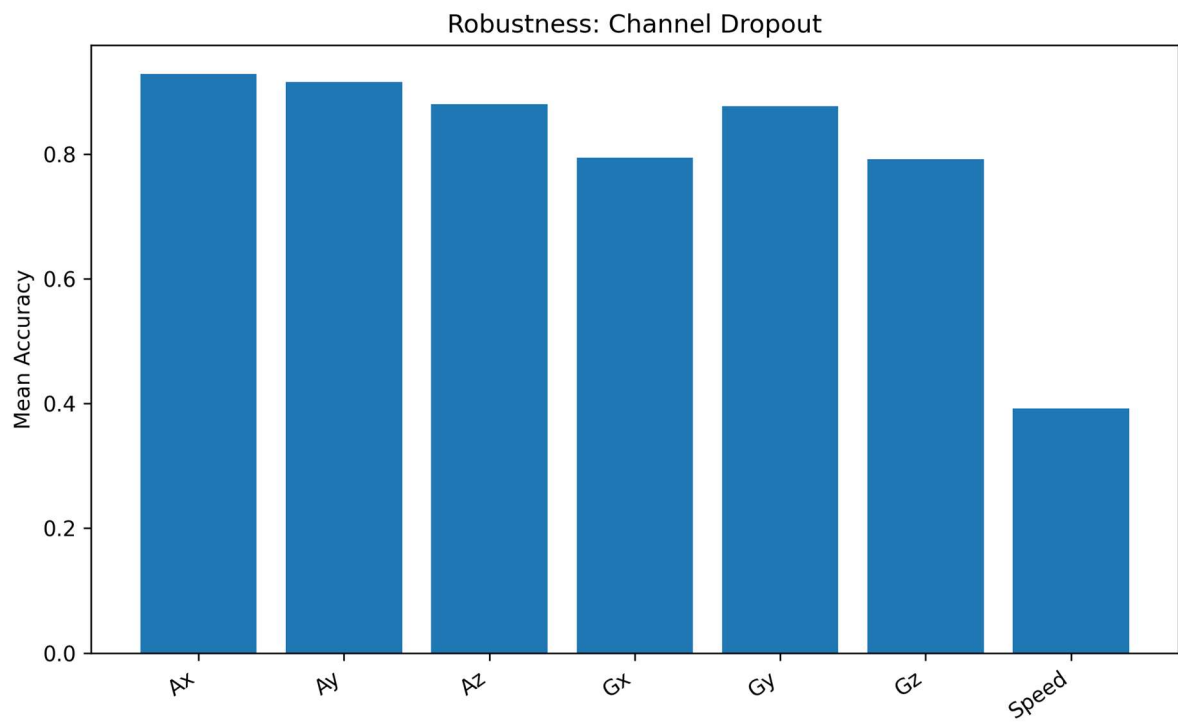
Confusion matrix: xgboost

7. TCNCA Ablation and Robustness

Variant	Accuracy	Macro F1	Latency (ms)	Parameters
tcnca_v1	0.6411	0.5202	0.2558	0
tcnca_v2	0.6364	0.5217	0.3135	0
tcnca_v3	0.7081	0.6858	0.3180	0

Robustness Channel Dropout

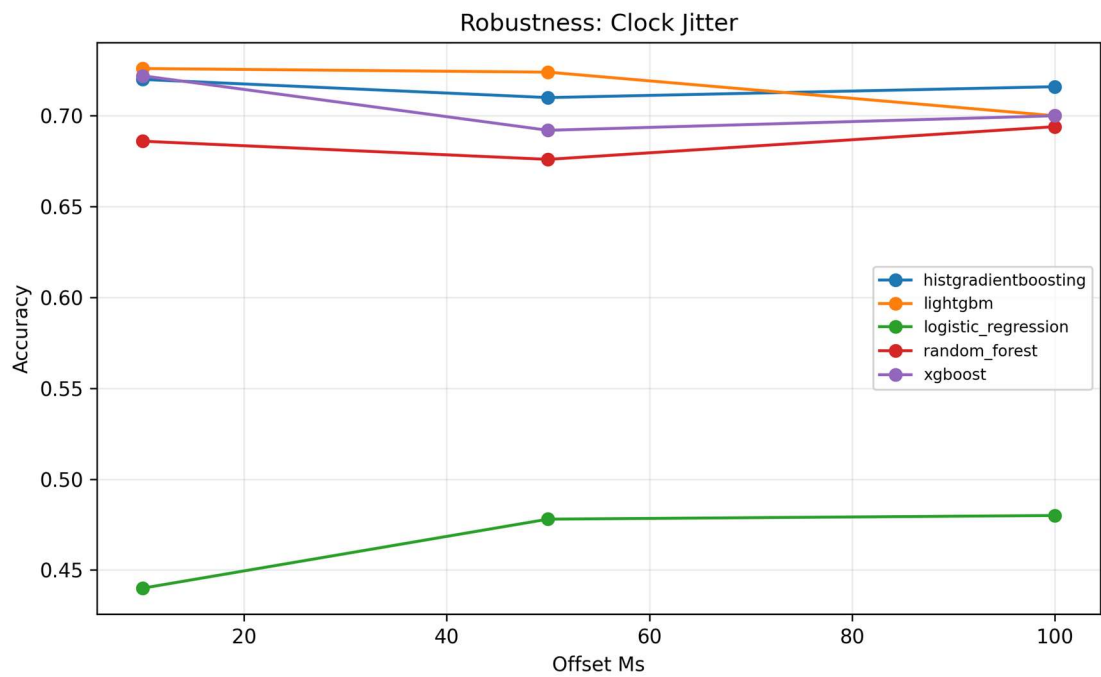
model	channel	accuracy	macro_f1
histgradientboosting	Ax	0.9440	0.9431
lightgbm	Ax	0.9420	0.9435
random_forest	Ax	0.9400	0.9411
xgboost	Ax	0.9440	0.9461
logistic_regression	Ax	0.8720	0.8592
histgradientboosting	Ay	0.9360	0.9409
lightgbm	Ay	0.9360	0.9401
random_forest	Ay	0.9280	0.9225
xgboost	Ay	0.9360	0.9389
logistic_regression	Ay	0.8420	0.8448
histgradientboosting	Az	0.9320	0.9329
lightgbm	Az	0.9360	0.9386
random_forest	Az	0.9040	0.8973
xgboost	Az	0.9300	0.9251
logistic_regression	Az	0.7000	0.7113
histgradientboosting	Gx	0.8580	0.7740
lightgbm	Gx	0.8500	0.7767
random_forest	Gx	0.9360	0.9359
xgboost	Gx	0.8620	0.7882
logistic_regression	Gx	0.4640	0.4246
histgradientboosting	Gy	0.9120	0.9177
lightgbm	Gy	0.9160	0.9241
random_forest	Gy	0.9380	0.9401
xgboost	Gy	0.9180	0.9265
logistic_regression	Gy	0.7020	0.7245
histgradientboosting	Gz	0.8200	0.5889
lightgbm	Gz	0.8200	0.5955
random_forest	Gz	0.7540	0.4950
xgboost	Gz	0.8040	0.5375
logistic_regression	Gz	0.7600	0.6303
histgradientboosting	Speed	0.4520	0.4959
lightgbm	Speed	0.5360	0.5372
random_forest	Speed	0.1500	0.0881
xgboost	Speed	0.5480	0.5990
logistic_regression	Speed	0.2760	0.4018



Robustness Channel Dropout summary

Robustness Clock Jitter

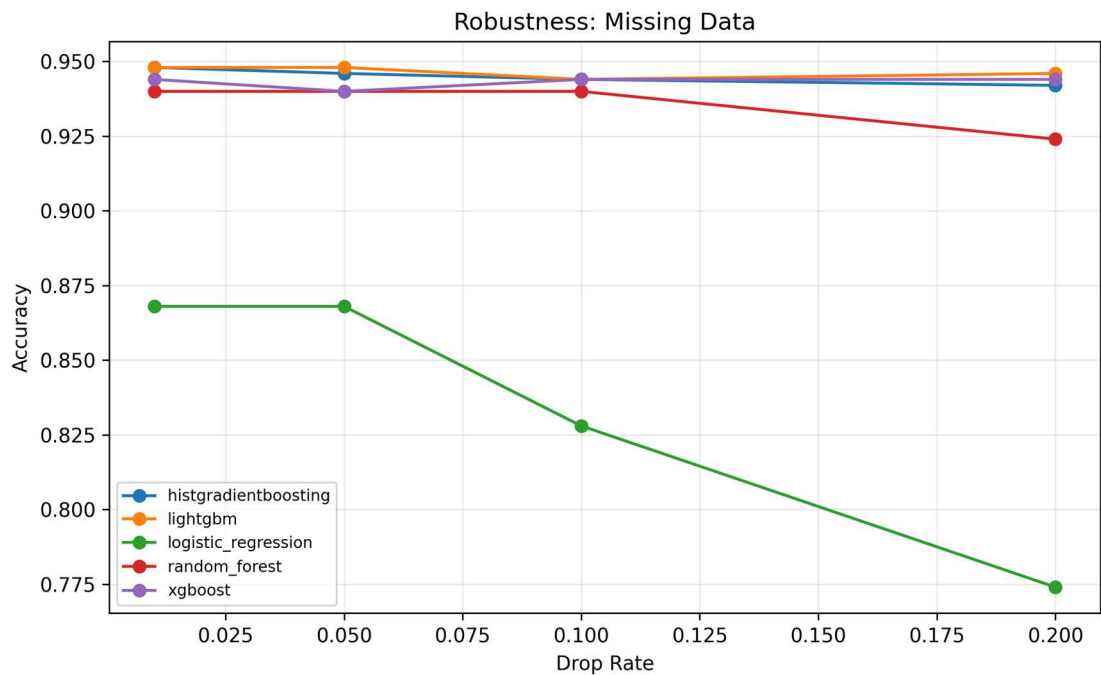
model	offset_ms	accuracy	macro_f1
histgradientboosting	10	0.7200	0.6941
lightgbm	10	0.7260	0.7040
random_forest	10	0.6860	0.6571
xgboost	10	0.7220	0.6772
logistic_regression	10	0.4400	0.4518
histgradientboosting	50	0.7100	0.6656
lightgbm	50	0.7240	0.6879
random_forest	50	0.6760	0.6389
xgboost	50	0.6920	0.6395
logistic_regression	50	0.4780	0.4774
histgradientboosting	100	0.7160	0.6850
lightgbm	100	0.7000	0.6752
random_forest	100	0.6940	0.6764
xgboost	100	0.7000	0.6718
logistic_regression	100	0.4800	0.4672



Robustness Clock Jitter summary

Robustness Missing

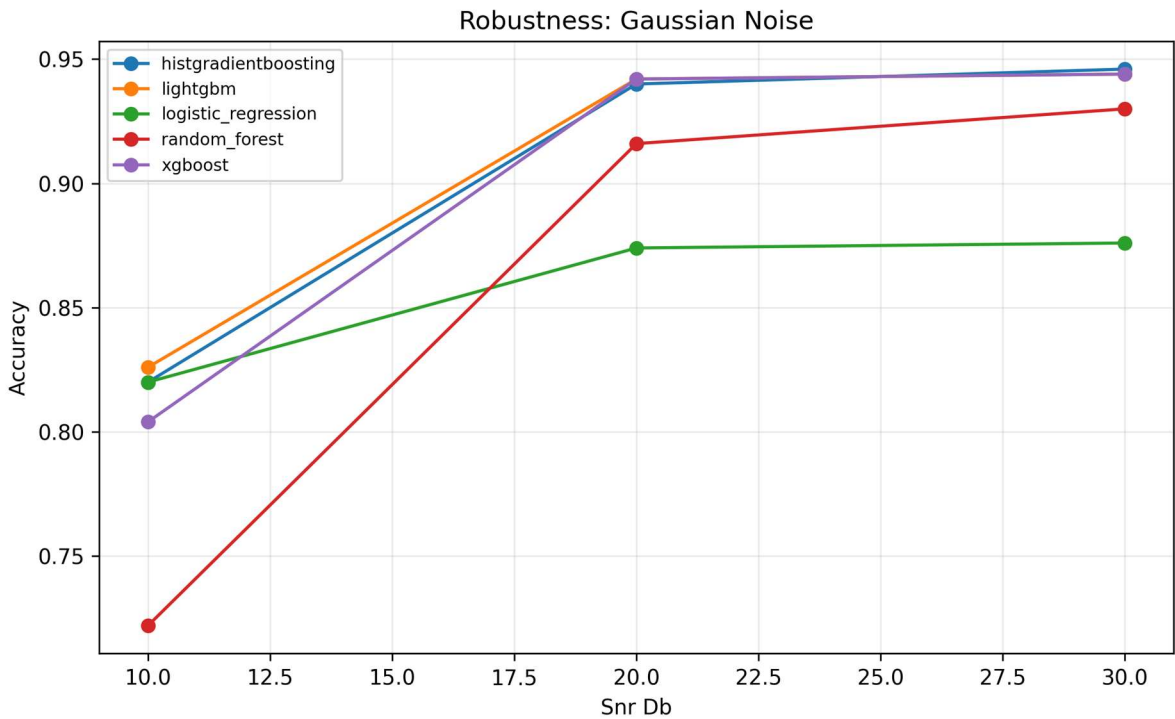
model	drop_rate	accuracy	macro_f1
histgradientboosting	0.0100	0.9480	0.9472
lightgbm	0.0100	0.9480	0.9480
random_forest	0.0100	0.9400	0.9427
xgboost	0.0100	0.9440	0.9458
logistic_regression	0.0100	0.8680	0.8558
histgradientboosting	0.0500	0.9460	0.9461
lightgbm	0.0500	0.9480	0.9466
random_forest	0.0500	0.9400	0.9413
xgboost	0.0500	0.9400	0.9436
logistic_regression	0.0500	0.8680	0.8662
histgradientboosting	0.1000	0.9440	0.9474
lightgbm	0.1000	0.9440	0.9425
random_forest	0.1000	0.9400	0.9416
xgboost	0.1000	0.9440	0.9485
logistic_regression	0.1000	0.8280	0.8314
histgradientboosting	0.2000	0.9420	0.9401
lightgbm	0.2000	0.9460	0.9481
random_forest	0.2000	0.9240	0.9200
xgboost	0.2000	0.9440	0.9461
logistic_regression	0.2000	0.7740	0.7778



Robustness Missing summary

Robustness Noise

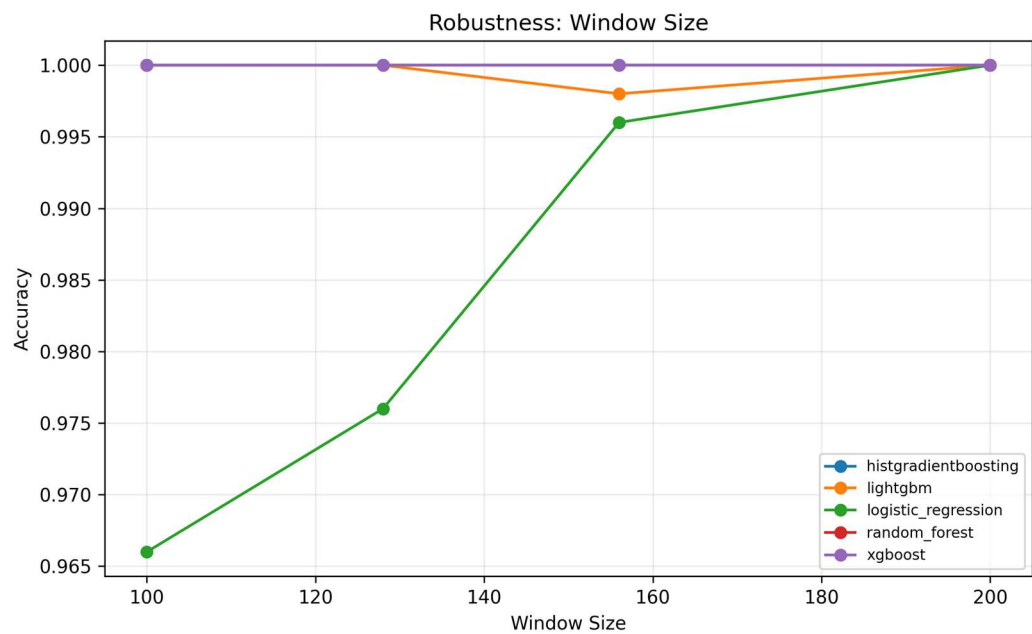
model	snr_db	accuracy	macro_f1
histgradientboosting	30	0.9460	0.9461
lightgbm	30	0.9440	0.9443
random_forest	30	0.9300	0.9297
xgboost	30	0.9440	0.9485
logistic_regression	30	0.8760	0.8642
histgradientboosting	20	0.9400	0.9373
lightgbm	20	0.9420	0.9404
random_forest	20	0.9160	0.9139
xgboost	20	0.9420	0.9419
logistic_regression	20	0.8740	0.8634
histgradientboosting	10	0.8200	0.8282
lightgbm	10	0.8260	0.8424
random_forest	10	0.7220	0.7109
xgboost	10	0.8040	0.7957
logistic_regression	10	0.8200	0.8287



Robustness Noise summary

Robustness Window Size

model	window_size	accuracy	macro_f1
histgradientboosting	100	1.0000	1.0000
lightgbm	100	1.0000	1.0000
random_forest	100	1.0000	1.0000
xgboost	100	1.0000	1.0000
logistic_regression	100	0.9660	0.9708
histgradientboosting	128	1.0000	1.0000
lightgbm	128	1.0000	1.0000
random_forest	128	1.0000	1.0000
xgboost	128	1.0000	1.0000
logistic_regression	128	0.9760	0.9826
histgradientboosting	156	1.0000	1.0000
lightgbm	156	0.9980	0.9973
random_forest	156	1.0000	1.0000
xgboost	156	1.0000	1.0000
logistic_regression	156	0.9960	0.9978
histgradientboosting	200	1.0000	1.0000
lightgbm	200	1.0000	1.0000
random_forest	200	1.0000	1.0000
xgboost	200	1.0000	1.0000
logistic_regression	200	1.0000	1.0000



Robustness Window Size summary

8. Final Comparison and Recommendation

Model	Accuracy	Macro F1	MAE	P95 latency (ms)
histgradientboosting	0.9460	0.9461	0.1680	116.5344
lightgbm	0.9440	0.9443	0.1680	5.9961
random_forest	0.9380	0.9415	0.1940	47.5854
xgboost	0.9400	0.9412	0.1860	4.7093
logistic_regression	0.8740	0.8630	0.3520	0.2273
linear_svm	0.8440	0.8280	0.3980	0.1302
mlp_tabular	0.7480	0.7363	0.8000	272.5350
conv_transformer	0.7520	0.7274	0.7520	1.4676
gru	0.7140	0.6828	0.9000	356.3285
lstm	0.7260	0.6504	0.7920	253.6587
fusformer	0.6740	0.6464	1.0500	0.7482
mega	0.6620	0.6260	1.0300	0.8412
convnet_har	0.6820	0.6023	0.9940	1.2880
tcnca_v3	0.6040	0.5639	1.2500	0.3501
rbf_svm	0.5940	0.5635	1.2520	1.5168
tcnca_v2	0.5860	0.4746	1.2720	0.5039
tcnca_v1	0.5760	0.4444	1.3340	0.4615
tst	0.5400	0.3941	1.5020	0.8614
s4d	0.4980	0.2541	1.6120	4.0810
s5_onnx	0.4980	0.2541	1.6120	3.9633
lru	0.4980	0.2541	1.6120	4.4430
sgconv	0.4820	0.1301	1.6160	1.1704
resnet1d	0.4820	0.1301	1.6160	0.3122
egodrive_rt	0.2880	0.0894	2.3840	0.0050
egodrive_max	0.2880	0.0894	2.3840	0.0046

Recommendation: use XGBoost as the production classification baseline because it combines 94.00% accuracy, strong macro F1, and low measured latency.

HistGradientBoosting has the highest score in the 500-window comparison, but its p95 latency is materially higher. ConvTransformer is the strongest compact neural candidate among the measured transformer models. Final deployment selection requires Raspberry Pi 5 measurements.

9. Limitations

- Only two sessions are available; generalisation to unseen rides or riders cannot be claimed.
- The unified 500-window benchmark uses the full window artifact because the random test split has only 209 windows, so training-data reuse is possible.
- Session-wise comparisons reuse existing random-split-trained artifacts and are diagnostic rather than clean retrain-per-session validation.
- MAE on encoded labels is not semantically ordered and should not replace macro F1.
- Clock-jitter experiments are simulated, not real field clock drift measurements.
- EgoDriveRT weights are unavailable and EgoDriveMax is rejected for deployment constraint.